

HealthConnect Clinic — Week 8 Final Project Summary

Intern Name:	Grai Rudolf
Role & Track:	Junior Machine Learning Engineer MLE Track
Project:	HealthConnect Clinic — Final Integration Showcase
Status:	Demonstration & Integration Ready (100% Tests Passing)

Project Background & Objectives

HealthConnect Clinic sought an intelligent, data-driven approach to mitigate high patient appointment no-show rates (~48.5%). Over the 5-week lab journey (Weeks 4–8), the ML Engineering track designed and delivered a modular, reliable prediction and clinical intervention workflow.

Summary of Key Achievements

- **Zero Data Leakage:** Strictly isolated post-arrival features (waiting_time_minutes) and implemented patient-level grouped splitting (GroupShuffleSplit/GroupKFold) across 1,680 unique patients.
- **Cost-Sensitive High Recall:** Tuned decision threshold to 0.35 in alignment with Data Science handoff, capturing 94.2% of no-shows on unseen patient test splits (ROC-AUC = 0.6846).
- **Clinical Risk Tier Engine:** Categorized appointments into High Risk (WhatsApp + buffer slots), Moderate Risk (SMS reminders), and Low Risk (standard email).
- **Seamless HC-POD Integration:** Established bi-directional contract with Data Science (Mairame Samba Niang), providing schema validation and accepting serialized model artifacts.
- **Exhaustive Quality Assurance:** 21 automated tests (Unit, Integration, Reliability, Contract) all passing cleanly.

Final Performance & Handoff Summary

Component	Specification / Result	Operational Purpose
Model Algorithm	Random Forest (150 trees, max_depth=8)	Ensemble robustness and non-linear feature capture
ROC-AUC	0.6846 (Unseen Patients)	Outperforms linear baseline (+1.11%)
Recall on No-Shows	94.20% (Threshold = 0.35)	Minimizes empty doctor schedules
Demographic Fairness	Female AUC 0.6966 Male AUC 0.6719	Documented fairness gap disclosed transparently
Input / Output Format	12-feature raw dict -> JSON Risk Action	Fast microservice & GenAI assistant integration

HC-POD Collaboration & Final Readiness

Cross-track handoffs with Data Science, Data Analytics, Generative AI, and Project Management were verified through automated contract tests and end-to-end inference execution. The ML Engineering contribution is fully documented, container-ready, and prepared for final presentation.